

Chapter 1

An Overview of Computer and Microprocessor

Google Classroom

- Class Name: MPCA
- Class code: zqib5ai

Overview: 2 hrs

1. Brief overview of Microprocessor and Microcontroller
2. Organization of Computer systems
3. Architecture of Computer
4. Applications of Microprocessor and Microcontroller

1. Brief overview of Microprocessor and Microcontroller

- Differences between:
 - Microcomputer – a computer with a microprocessor as its CPU. Includes memory, I/O etc.
 - Microprocessor – silicon chip which includes ALU, register circuits & control circuits
 - Microcontroller – silicon chip which includes microprocessor, memory & I/O in a single package.

Microprocessor

- The word comes from the combination **micro** and **processor**.
 - Processor means a device that processes whatever. In this context processor means a device that processes numbers, specifically binary numbers, 0's and 1's.
 - To process means to manipulate. It is a general term that describes all manipulation. Again in this content, it means to perform certain operations on the numbers that depend on the microprocessor's design.

Definition of the Microprocessor

- The microprocessor is a programmable device that
 - takes in numbers,
 - performs on them arithmetic or logical operations according to the program stored in memory
 - and then produces other numbers as a result.

- **Takes in:** The data that the microprocessor manipulates must come from somewhere.
 - It comes from what is called “input devices”.
 - These are devices that bring data into the system from the outside world.
 - These represent devices such as a keyboard, a mouse, switches, and the like.

- – **Numbers**: The microprocessor has a very narrow view on life. It only understands binary numbers.
- A binary digit is called a bit (which comes from **binary digit**).
- The microprocessor recognizes and processes a group of bits together. This group of bits is called a “word”.
- The number of bits in a Microprocessor’s word, is a measure of its “abilities”.

– Words, Bytes, etc.

- The earliest microprocessor (the Intel 8088 and Motorola's 6800) recognized 8-bit words.
 - They processed information 8-bits at a time. That's why they are called "8-bit processors". They can handle large numbers, but in order to process these numbers, they broke them into 8-bit pieces and processed each group of 8-bits separately.
- Later microprocessors (8086 and 68000) were designed with 16-bit words.
 - A group of 8-bits were referred to as a "half-word" or "byte".
 - A group of 4 bits is called a "nibble".
 - Also, 32 bit groups were given the name "long word".
- Today, all processors manipulate at least 32 bits at a time and there exists microprocessors that can process 64, 80, 128 bits

– Arithmetic and Logic Operations:

- Every microprocessor has arithmetic operations such as add and subtract as part of its instruction set.
 - Most microprocessors will have operations such as multiply and divide.
 - Some of the newer ones will have complex operations such as square root.
- In addition, microprocessors have logic operations as well. Such as AND, OR, XOR, shift left, shift right, etc.
- Again, the number and types of operations define the microprocessor's instruction set and depends on the specific microprocessor.

– Stored in memory :

- First, what is memory?
 - Memory is the location where information is kept while not in current use.
 - Memory is a collection of storage devices. Usually, each storage device holds one bit. Also, in most kinds of memory, these storage devices are grouped into groups of 8. These 8 storage locations can only be accessed together. So, one can only read or write in terms of bytes to and from memory.
 - Memory is usually measured by the number of bytes it can hold.
- It is measured in Kilos, Megas and lately Gigas. A Kilo in computer language is $2^{10}=1024$. So, a KB (KiloByte) is 1024 bytes. Mega is 1024 Kilos and Giga is 1024 Mega.

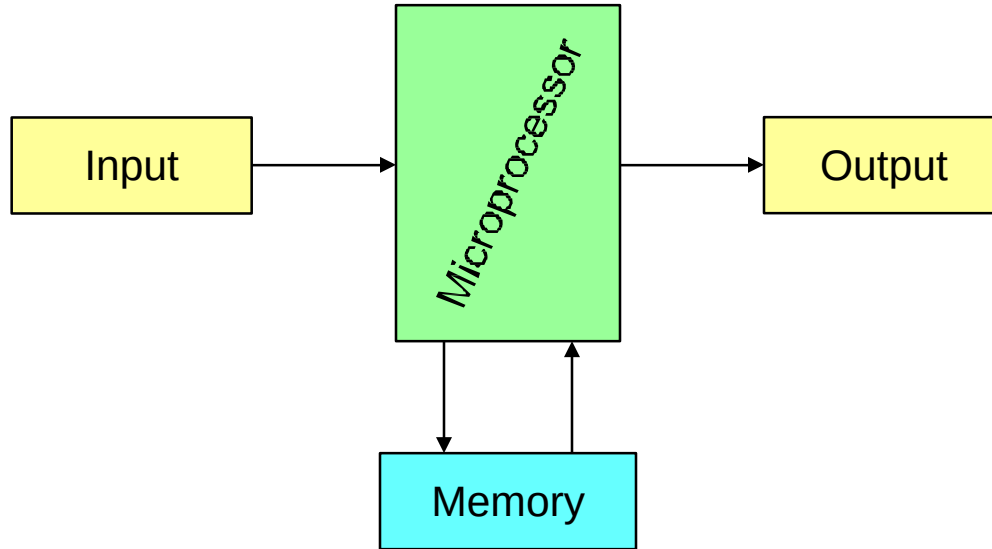
– Stored in memory:

- When a program is entered into a computer, it is stored in memory. Then as the microprocessor starts to execute the instructions, it brings the instructions from memory one at a time.
- Memory is also used to hold the data.
 - – The microprocessor reads (brings in) the data from memory when it needs it and writes (stores) the results into memory when it is done.

- **Produces:** For the user to see the result of the execution of the program, the results must be presented in a human readable form.
 - The results must be presented on an output device.
 - This can be the monitor, a paper from the printer, a simple LED or many other forms.

A Microprocessor-based system

From the above description, we can draw the following block diagram to represent a microprocessor-based system:

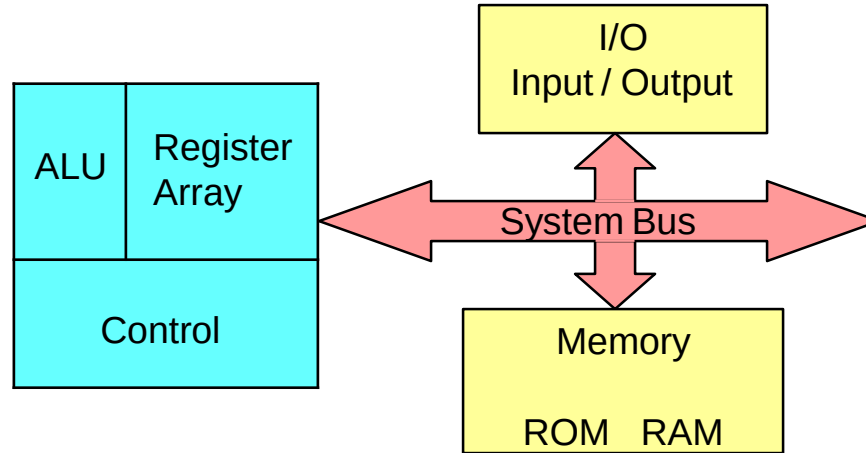


Inside The Microprocessor

- Internally, the microprocessor is made up of 3 main units.
 - The Arithmetic/Logic Unit (ALU)
 - The Control Unit.
 - An array of registers for holding data while it is being manipulated.

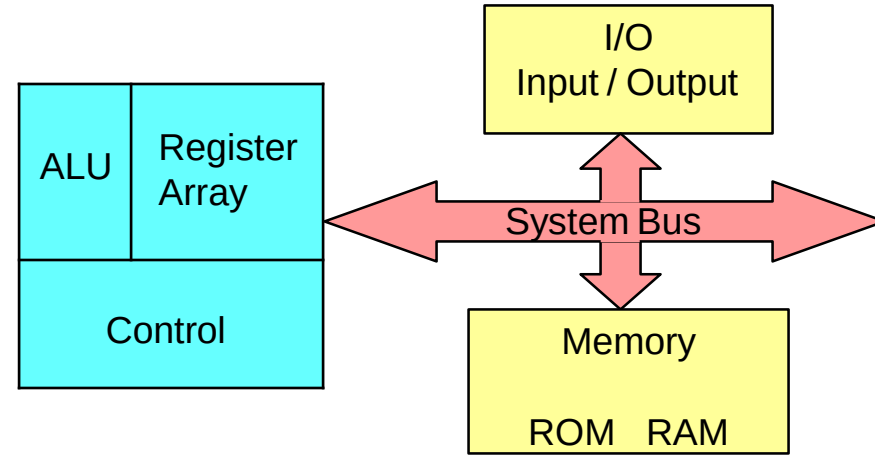
Organization of a microprocessor- based system

- Let's expand the picture a bit.



Memory

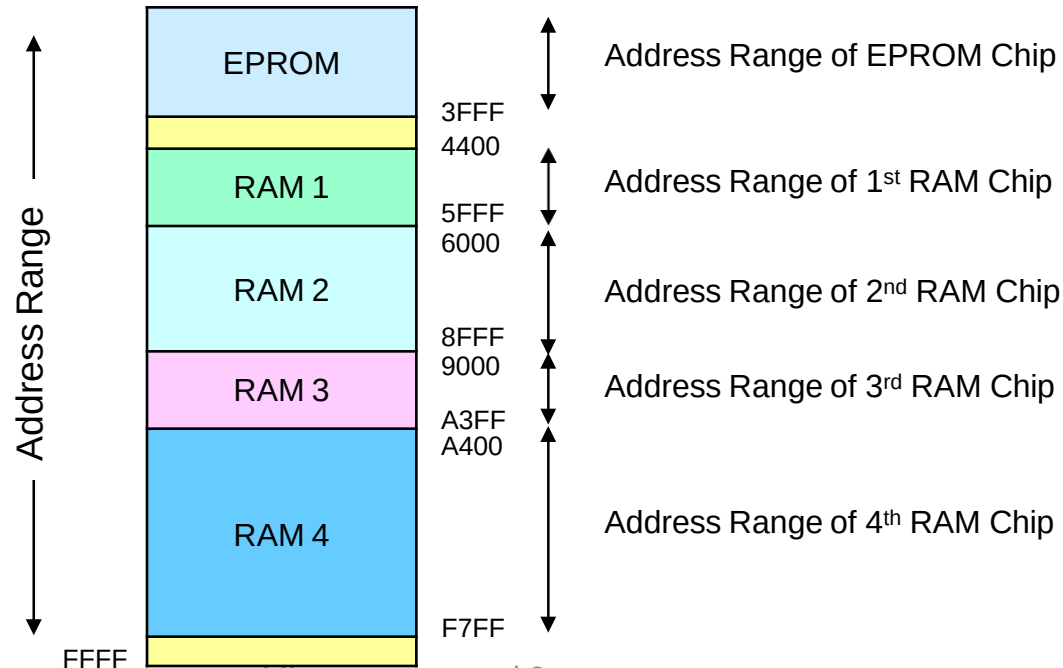
- Memory stores information such as instructions and data in binary format (0 and 1).
- It provides this information to the microprocessor whenever it is needed.
- Usually, there is a memory “sub-system” in a microprocessor-based system. This sub-system includes:
 - The registers inside the microprocessor
 - Read Only Memory (ROM)
 - used to store information that does not change.
 - Random Access Memory (RAM) (also known as Read/Write Memory).
 - used to store information supplied by the user. Such as programs and data.



- To execute a program:
 - the user enters its instructions in binary format into the memory.
 - The microprocessor then reads these instructions and whatever data is needed from memory, executes the instructions and places the results either in memory or produces it on an output device.

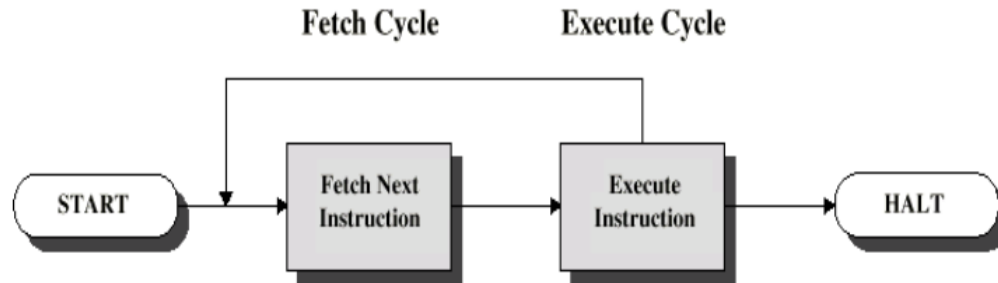
Memory Map and Addresses

- The memory map is a picture representation of the address range and shows where the different memory chips are located within the address range.



The three cycle instruction execution model

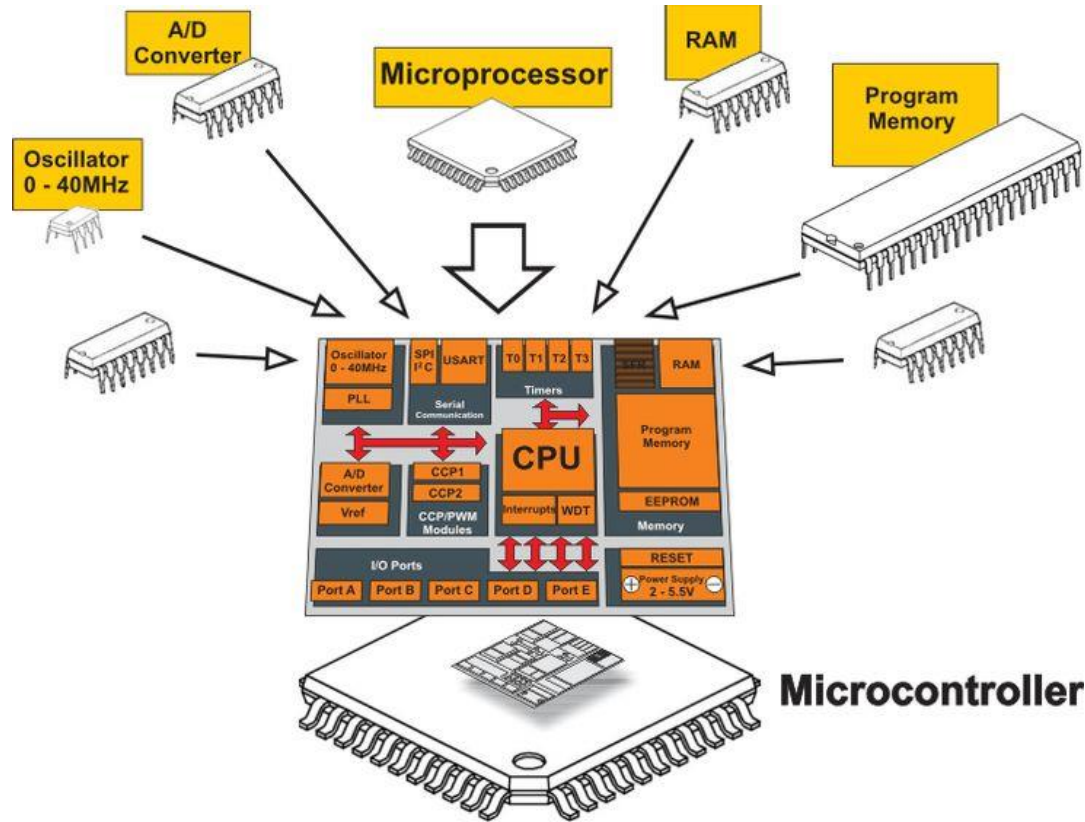
- To execute a program, the microprocessor “reads” each instruction from memory, “interprets” it, then “executes” it.
- To use the right names for the cycles:
 - The microprocessor **fetches** each instruction,
 - **decodes** it,
 - Then **executes** it.
- This sequence is continued until all instructions are performed.



Microcontroller

- A **microcontroller** is a chip optimized to control electronic devices.
- It is stored in a single integrated circuit which is dedicated to performing a particular task and execute one specific application.
- It is specially designed circuits for embedded applications and is widely used in automatically controlled electronic devices.
- It contains memory, processor, and programmable I/O.
- Eg: 8051 Microcontroller





Applications of Microprocessor and Microcontroller

Application of Microprocessor

1. **Computers:** Microprocessors form the core of computers, including personal computers, laptops, and servers. They handle tasks like executing instructions, managing memory, and controlling input/output devices.
2. **Embedded Systems:** Microprocessors are extensively used in embedded systems, which are specialized computer systems designed for specific applications. They can be found in appliances, automotive systems, industrial control systems, and consumer electronics.
3. **Communication Systems:** Microprocessors are used in communication systems like routers, modems, and network switches. They handle data transmission, network protocols, and routing algorithms.
4. **Automotive Industry:** Microprocessors play a vital role in modern vehicles. They control functions such as engine management, fuel injection systems, anti-lock braking systems (ABS), airbag control, entertainment systems, and navigation systems.
5. **Robotics:** Microprocessors are used for controlling the movements and operations of robots. They process sensor data, execute control algorithms, and enable autonomous decision-making.

Application of Microprocessor

6. **Aerospace and Defense:** Microprocessors are essential components in aerospace and defense systems, including aircraft flight controls, missile guidance systems, satellite communications, and radar systems.
7. **Consumer Electronics:** Microprocessors are widely used in consumer electronics products like smart TVs, gaming consoles, digital cameras, audio systems, and wearable devices. They provide the necessary processing power for multimedia applications and user interfaces.
8. **Internet of Things (IoT):** Microprocessors play a key role in IoT devices, enabling connectivity, data processing, and control in applications such as smart homes, industrial automation, agriculture monitoring, and environmental sensing.
9. **Medical Devices:** Microprocessors are utilized in medical equipment and devices, including patient monitoring systems, imaging devices (e.g., MRI and CT scanners), implantable devices, and laboratory equipment. They enable data processing, control, and real-time analysis.
10. **Industrial Control Systems:** Microprocessors are employed in industrial automation and control systems to monitor and control manufacturing processes, robotics, assembly lines, and supervisory control and data acquisition (SCADA) systems.

Application of Microcontroller

1. **Home Automation:** Microcontrollers are used to control and automate various functions in smart homes, including lighting systems, temperature control, security systems, and appliances.
2. **Internet of Things (IoT):** Microcontrollers play a crucial role in IoT devices by providing connectivity, data processing, and control capabilities. They are used in applications such as environmental monitoring, smart agriculture, and industrial automation.
3. **Automotive Systems:** Microcontrollers are extensively used in automotive applications, including engine control units (ECUs), anti-lock braking systems (ABS), airbag control, dashboard displays, and entertainment systems.
4. **Consumer Electronics:** Microcontrollers are found in a wide range of consumer electronics, such as remote controls, digital cameras, home entertainment systems, and wearable devices. They provide control and interface functionality.
5. **Medical Devices:** Microcontrollers are used in various medical devices, including patient monitoring systems, insulin pumps, prosthetic devices, and diagnostic equipment. They enable data processing, control, and communication functions.

Application of Microcontroller

6. **Industrial Automation:** Microcontrollers are used in industrial control systems to monitor and control processes, machinery, and equipment. They play a key role in programmable logic controllers (PLCs), robotics, and supervisory control and data acquisition (SCADA) systems.
7. **Robotics:** Microcontrollers are crucial for controlling the movements, sensors, and actuators of robots. They handle real-time data processing, control algorithms, and communication interfaces.
8. **Security Systems:** Microcontrollers are employed in security systems such as access control systems, alarm systems, and surveillance systems. They provide the necessary processing power and interfaces for monitoring and controlling security functions.
9. **Energy Management:** Microcontrollers are used in energy management systems for monitoring and controlling power consumption, renewable energy systems, and smart grid applications.
10. **Educational Tools:** Microcontrollers are widely used in educational settings for teaching and learning purposes. They provide a hands-on platform for students to learn about programming, electronics, and control systems.

Assignment:

- Differentiate between Microprocessor and Microcontroller

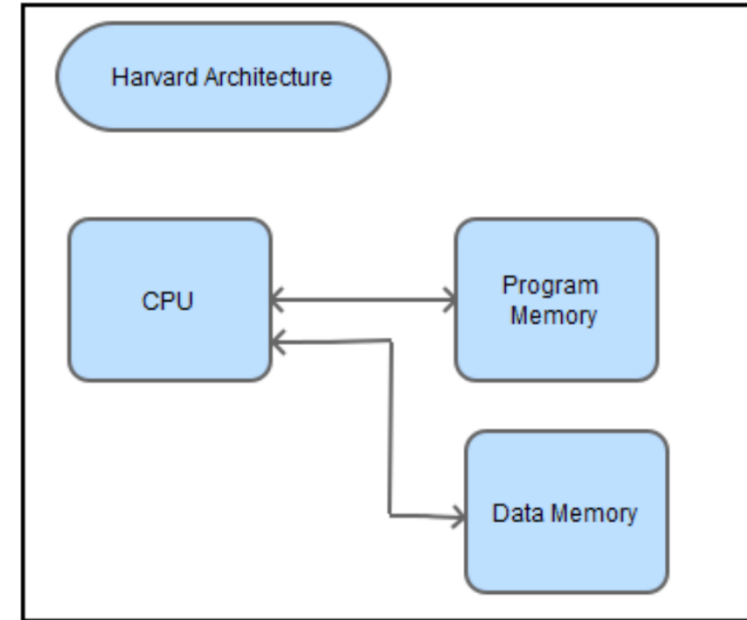
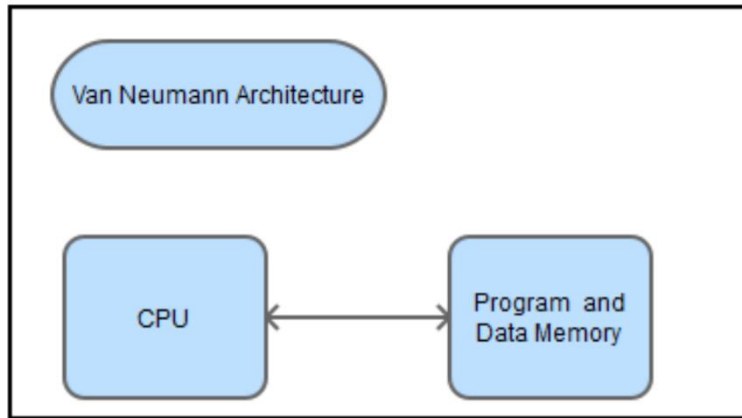
3. Architecture of Computer

Computer Architecture

- Computer architecture comprises rules, methods, and procedures that describe the execution and functionality of the entire computer system.
- In general terms, computer architecture refers to how a computer system is designed using compatible technologies.

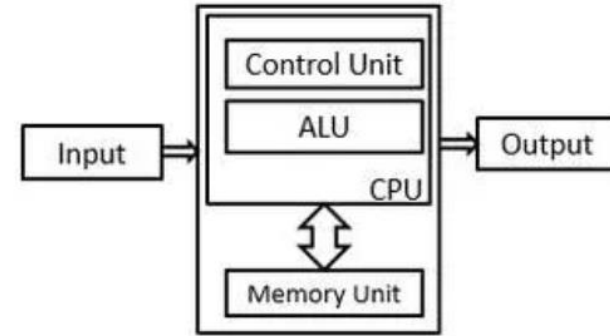
Types of Computer Architecture

- Von-Neumann Architecture
- Harvard Architecture



Von Neuman Architecture

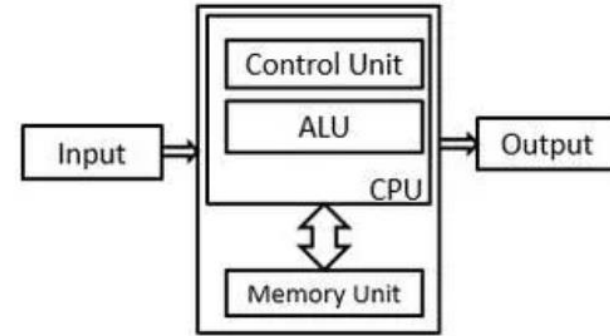
- The Von Neumann architecture is a type of computer architecture in which the central processing unit (CPU), memory, and input/output (I/O) devices all use a single shared bus for communication.
- It was named after John von Neumann, a pioneer in the development of electronic computers. The Von Neumann architecture is also known as the Von Neumann model or the Princeton architecture.
- In a Von Neumann architecture, instructions and data are stored in a single memory system. This memory is usually implemented as random access memory (RAM) and is connected to the CPU and I/O devices through a common bus. The CPU executes instructions by fetching them from memory, decoding them, and then executing them.



Von Neumann Model

Von Neuman Architecture

- The Von Neumann architecture consists of four main components:
 1. **CPU:** The central processing unit is the brain of the computer. It performs all the calculations and operations required to execute instructions.
 2. **Memory:** The memory holds both instructions and data that the CPU needs to execute. The memory is organized as a linear sequence of addressable cells, each containing a fixed number of bits.
 3. **Input/output (I/O) devices:** These devices are used to communicate with the outside world. Examples include keyboards, displays, and printers.
 4. **System bus:** The system bus is a collection of wires that connect the CPU, memory, and I/O devices. It is used to transmit data, instructions, and control signals between these components.



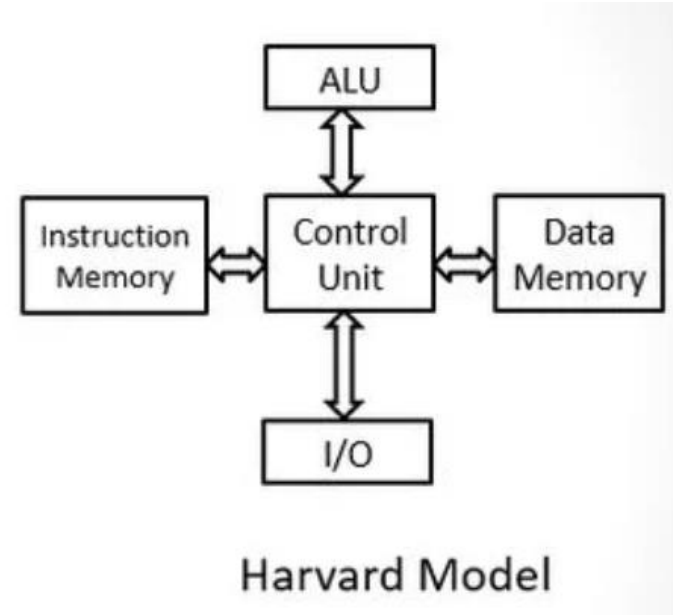
Von Neumann Model

Advantages / Disadvantages of Von-Neumann Architecture

1. Simple and flexible
 2. Less physical space is required than Harvard
 3. Handling just one memory block is simpler and easier to achieve
 4. Cheaper to use than Harvard
1. **Shared memory** - a defective program can overwrite another in memory, causing it to crash
 2. **Memory leaks** - some defective programs fail to release memory when they are finished with it, which could cause the computer to crash due to insufficient memory
 3. **Data bus speed** - the CPU is much faster than the data bus, meaning it often sits idle (Von Neumann bottleneck)
 4. **Fetch rate** - data and instructions share the same data bus, even though the rate at which each needs to be fetched is often very different

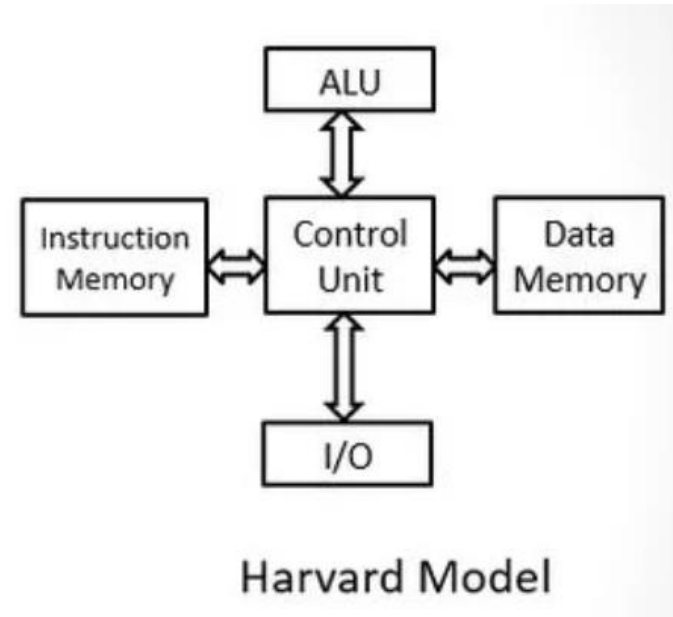
Harvard Architecture

- Harvard architecture is a type of computer architecture that has separate memory spaces for instructions and data.
- It was developed at Harvard University in the 1930s, and it is named after this institution.
- In a Harvard architecture system, the CPU accesses instruction and data memory spaces separately, which can lead to improved performance.



Harvard Architecture

- The Harvard architecture consists of the following main components:
 1. **CPU:** The central processing unit performs all the calculations and operations required to execute instructions.
 2. **Instruction memory:** This memory holds instructions that the CPU needs to execute. It is typically implemented as read-only memory (ROM) or flash memory.
 3. **Data memory:** This memory holds data that the CPU needs to perform computations. It is typically implemented as random access memory (RAM).
 4. **Input/output (I/O) devices:** These devices are used to communicate with the outside world. Examples include keyboards, displays, and printers.
 5. **System bus:** The system bus is a collection of wires that connect the CPU, instruction memory, data memory, and I/O devices. It is used to transmit data, instructions, and control signals between these components.



Advantages / Disadvantages of Harvard Architecture

1. **Fast and efficient data access:** Since Harvard architecture has separate memory spaces for instructions and data, it allows for parallel and simultaneous access to both memory spaces, which leads to faster and more efficient data access.
2. **Better performance:** The use of fixed instruction length, parallel processing, and optimized memory usage in Harvard architecture can lead to improved performance and faster execution of instructions.
3. **Suitable for real-time applications:** Harvard architecture is commonly used in embedded systems and other real-time applications where speed and efficiency are critical.
4. **Security:** The separation of instruction and data memory spaces can also provide a degree of security against certain types of attacks, such as buffer overflow attacks.

1. **Complexity:** The use of separate memory spaces for instructions and data in Harvard architecture adds to the complexity of the processor design and can increase the cost of manufacturing.
2. **Limited flexibility:** Harvard architecture has limited flexibility in terms of modifying instructions at runtime because instructions and data are stored in separate memory spaces. This can make certain types of programming more difficult or impossible to implement.
3. **Higher memory requirements:** Harvard architecture requires more memory than Von Neumann architecture, which can lead to higher costs and power consumption.
4. **Code size limitations:** Fixed instruction length in Harvard architecture can limit the size of code that can be executed, making it unsuitable for some applications with larger code bases.

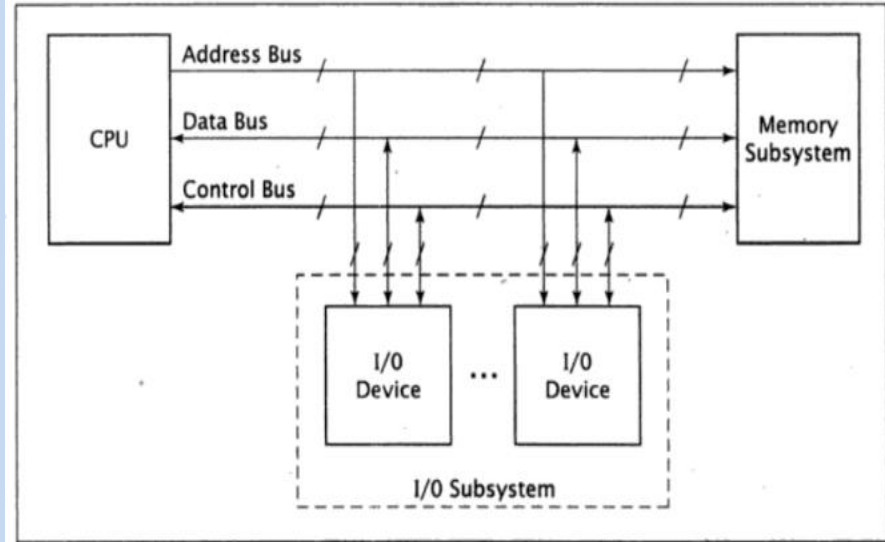
Assignment:

- Differentiate between Von Neumann and Harvard architectures

2. Organization of Computer systems

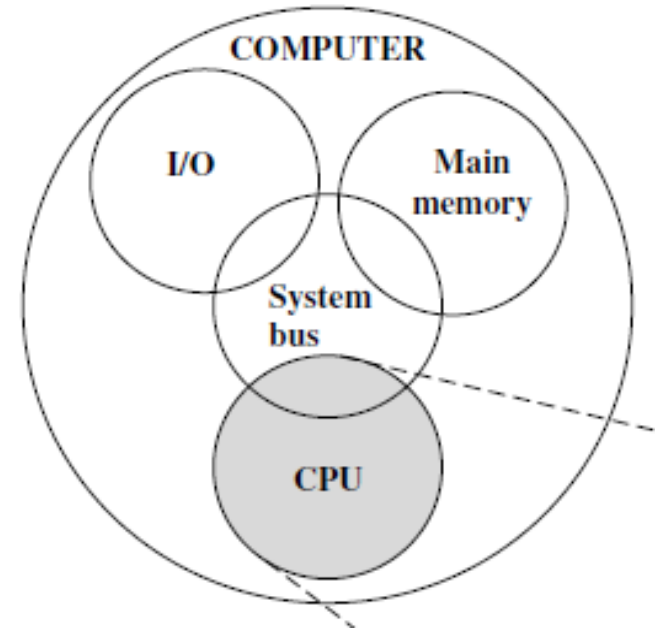
1. Basic Computer Organization

- This organization has three main components:
 - CPU,
 - memory subsystem, and
 - IO subsystem



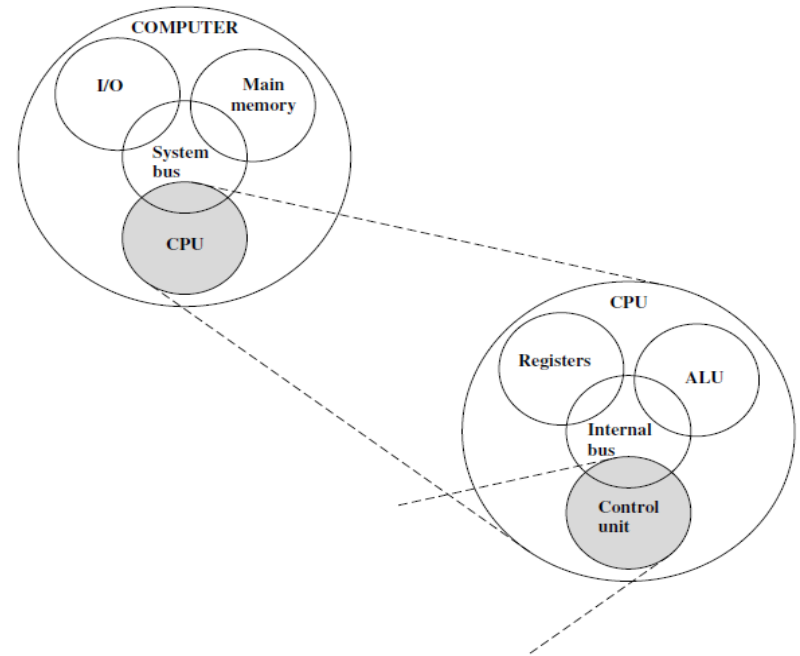
Structural view of Computer System

CPU	Controls the operation of the computer and performs its data processing functions; also referred as processor.
Main Memory	Stores data
I/O	Moves data between the memory and its external environment
System Interconnection	Some mechanism that provides for communication among CPU, main memory and I/O. it is done through system bus.



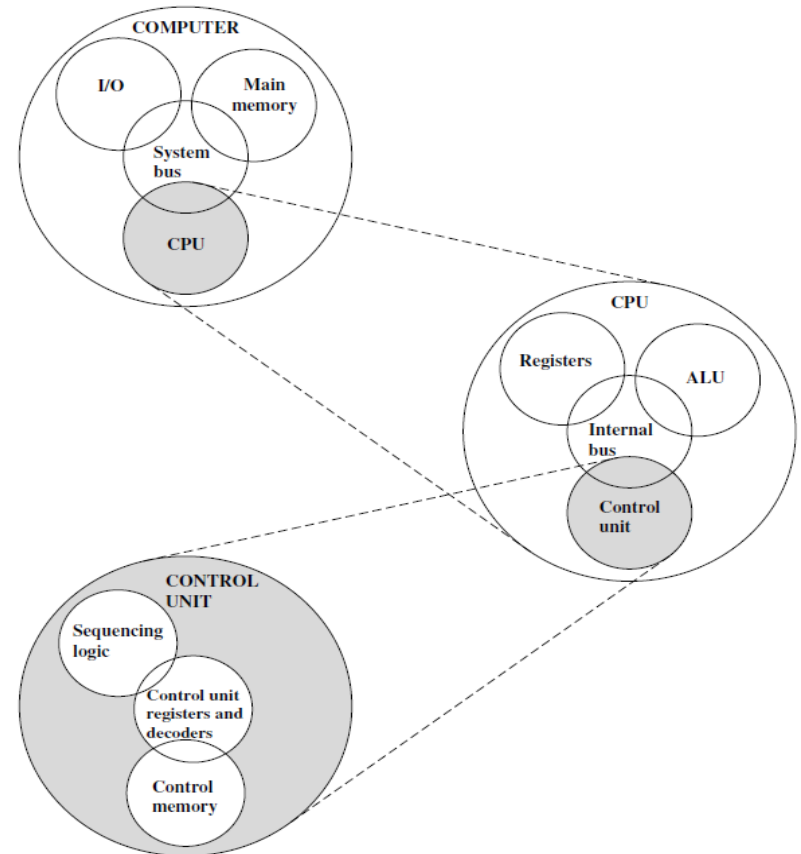
Structural view of Computer System (Contd.)

Control Unit:	Controls the operation of the CPU and hence the computer.
ALU:	Performs the computer's data processing function.
Registers:	Provides storage internal to the CPU.
CPU interconnection:	Some mechanisms that provide for communication among the CU, ALU and registers.



Structural view of Computer System (Contd.)

Control Memory:	The registers which store data related to control operation.
Sequential Logic:	Digital circuit whose output depends on current input and state of circuit.
CU register & decoder:	Performs storage and interpretation of the instruction.



2. System Buses

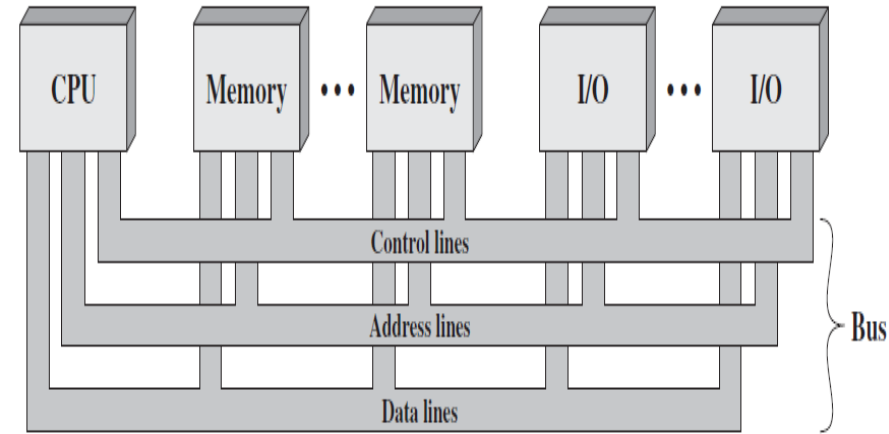
- Physically, a bus is a set of wires.
- The components of the computer are connected to the buses.
- To send information from one component to another, the source component outputs data onto the bus.
- The destination component then inputs this data from the bus.
- As the complexity of a computer system increases, it becomes more efficient at using buses rather than direct connections between every pair of devices.
- Buses use less space on a circuit board and require less power than a large number of direct connections.
- They also require fewer pins on the chip or chips that comprise the CPU.

2. System Buses (Contd.)

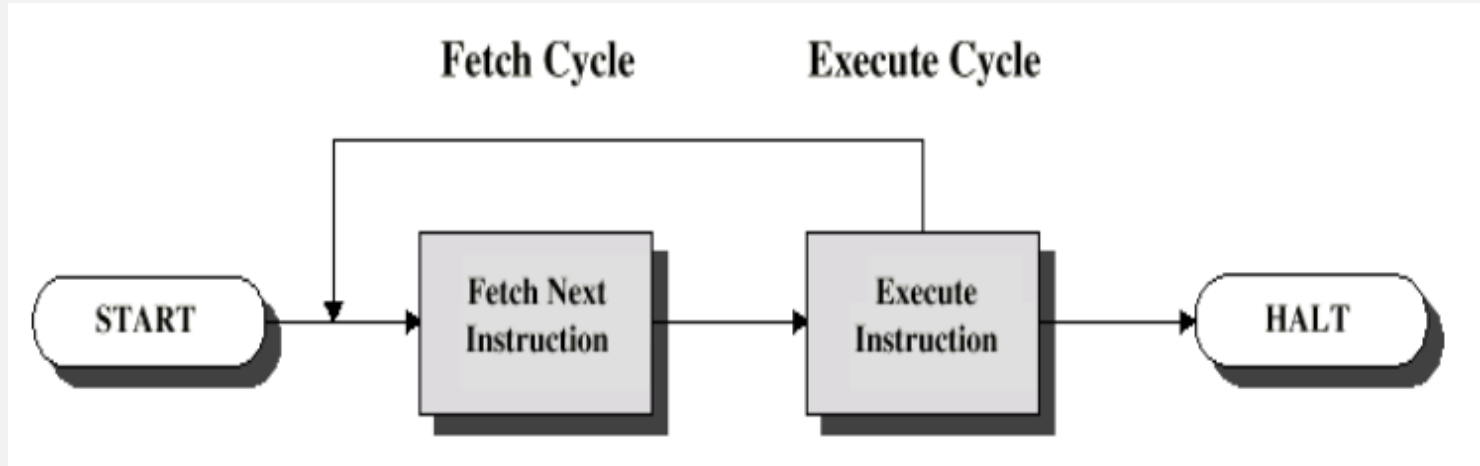
- It is a shared transmission medium. Multiple devices connect to the bus, and a signal transmitted by one device is available for reception by all other devices attached to the bus.
- If two devices transmit during the same time period, their signals will overlap and become garbled.
- Thus, only one device, at a time, can successfully transmit.
- Typically, a bus consists of multiple communication pathways or lines.
- Each line is capable of transmitting signals representing binary 1 or 0.
- An 8-bit unit of data can be transmitted over eight bus lines.
- A bus that connects major computer components (processor, memory, IO) is called a system bus.

2. System Buses (Contd.)

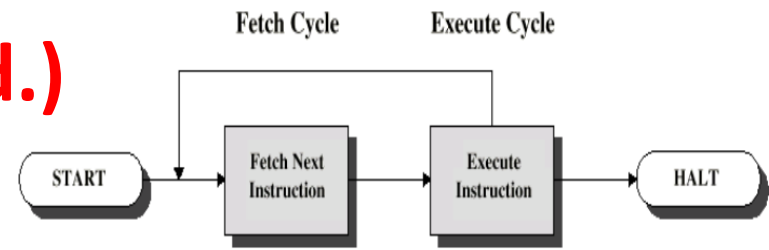
Data Lines	The data lines provide a path for moving data among system modules. These lines, collectively, are called the data bus.
Address Lines	The address lines are used to designate the source or destination of the data on the data bus.
Control Lines	The control lines are used to control the access to and the use of the data and address lines. Control signals transmit both command and timing information among system modules. • Timing signals indicate the validity of data and address information. • Command signals specify operation to be performed.



3. Instruction Cycle



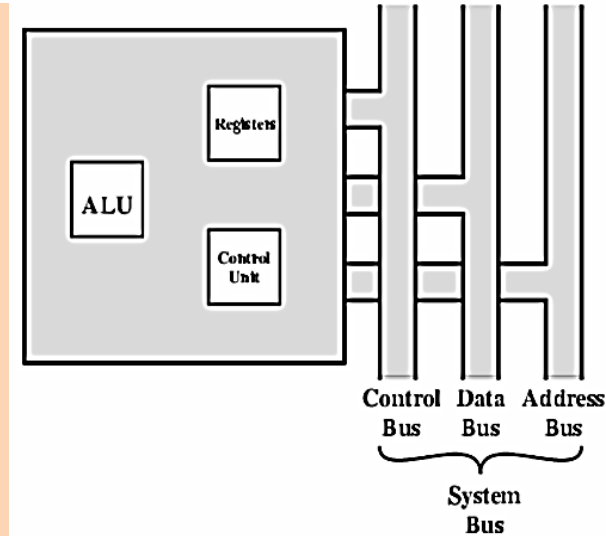
3. Instruction Cycle (Contd.)



- The instruction cycle is the procedure a microprocessor goes through to process an instruction.
- First the microprocessor **fetches** or reads, the instruction from memory.
- Then it **decodes** the instruction, determining which instruction it has fetched.
- Finally, it performs the operations necessary to **execute** the instruction.
- Each of these functions- fetch, decode and execute- consists of a sequence of one or more operations.

4. CPU organization

- CPU consists of three components:
 - ALU,
 - CU and
 - Registers.
- ALU performs arithmetic and logical operation,
- CU provides control signals, and
- Registers store binary information and provide very fast information to ALU and out of CPU.
- These components are connected with each other and with external environment through system bus as shown below:



2.4 Memory Sub-system Organization and Interfacing

- Memory is the group of circuits used to store data.
- Two types of Memory:
 1. RAM
 - a) SRAM:
 - More like registers, no need of refresh
 - b) DRAM:
 - Leaky capacitor, needs refresh
 2. ROM
 - a) PROM (Programmable ROM)
 - b) EPROM (Erasable PROM)
 - can be reprogrammed if required by exposing it to ultraviolet light.
 - c) EEPROM (Electrically Erasable PROM)
 - write and erase operations are performed one byte at a time , by application of electrical potential
 - d) Flash Memory

Reference: <https://www.guru99.com/different-types-ram-random-access-memory.html>
<https://www.scaler.com/topics/what-is-rom/>

RAM

Vs

ROM

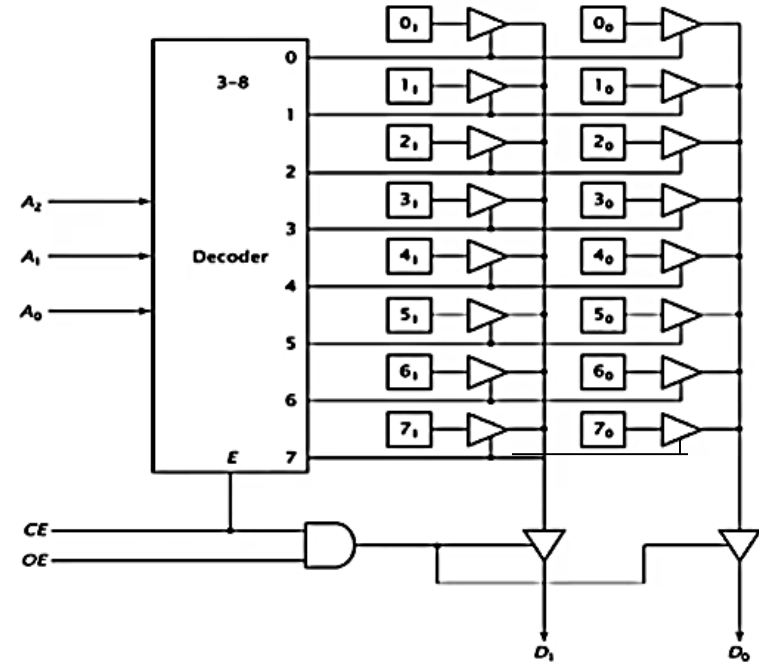
- Volatile in nature.
 - Used for reading and writing data in any order as required.
 - Data is stored and read many times to and from this type of memory.
 - Memory cells can be accessed for information transfer from any desired random location.
- Read only memory (ROM) contains a permanent pattern of data that cannot be changed.
 - A ROM is non-volatile, i.e., no power source is required to maintain the bit values in memory.
 - While it is possible to read a ROM, it is not possible to write new data into it.
 - The data or program is permanently presented in main memory and never be loaded from a secondary storage device with the advantage of ROM.

Internal Chip Organization

- Linear organization
- Two-dimensional organization

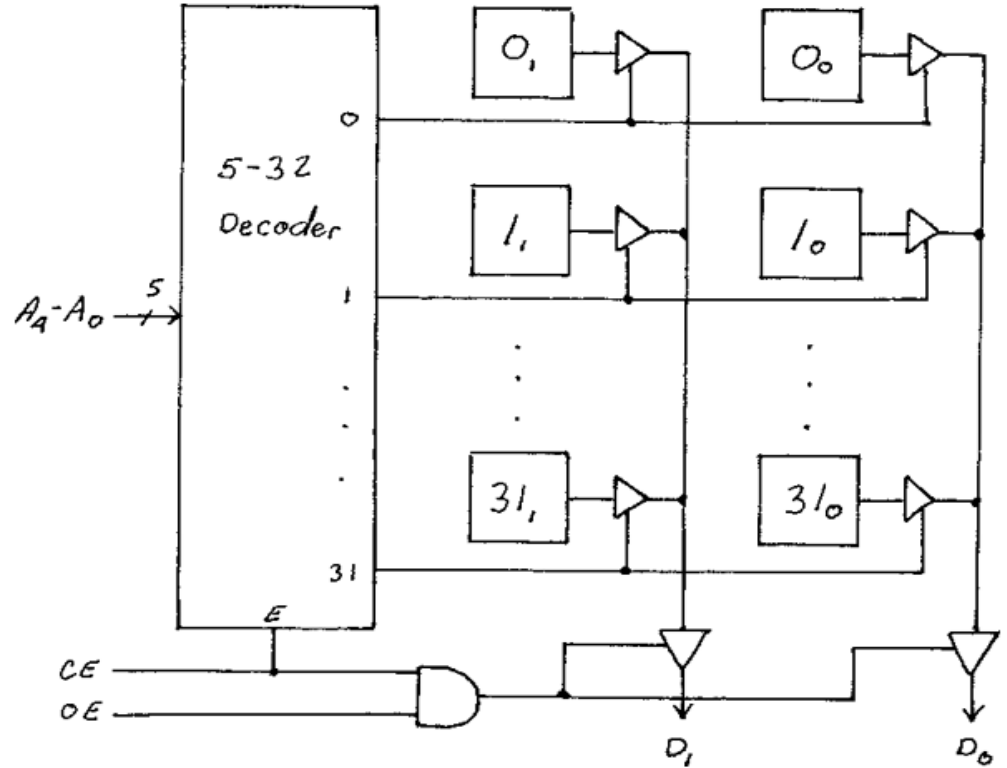
Linear organization

- Consider an 8 x 2 ROM chip: (Chip is in $2^n \times m$ representation where n is address line and m is data line)
 - Address line = 3
 - Data lines = 2
 - 16 bits of internal storage
 - If $CE = 0$, decoder is disabled and no location is selected. The Tri-state buffer for that location's cell are enabled, allowing data to pass to the output buffers.
 - If $CE = 1$ and $OE = 1$, tri-state buffer for that location cell is enabled and data is output from the chip; otherwise the outputs are tri-stated.



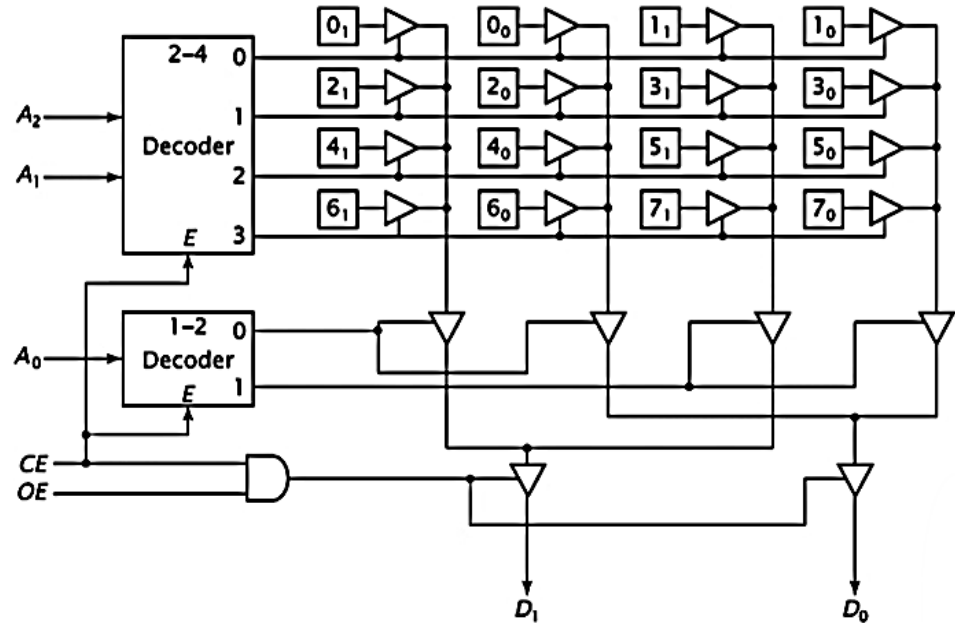
Classwork:

- show the internal linear configuration of a 32x2 memory chip.



Two-dimensional organization

- As the number of locations increases, the size of address decoder needed, becomes extremely large.
- To remedy this problem, the memory chip can be designed using multiple dimension of decoding.
- This configuration has four rows with four bits per row; each holds two data values.
- Here,
- $A_0 \rightarrow$ selects one of the 2 locations in a row
- $A_1, A_2 \rightarrow$ selects one of 4 rows.

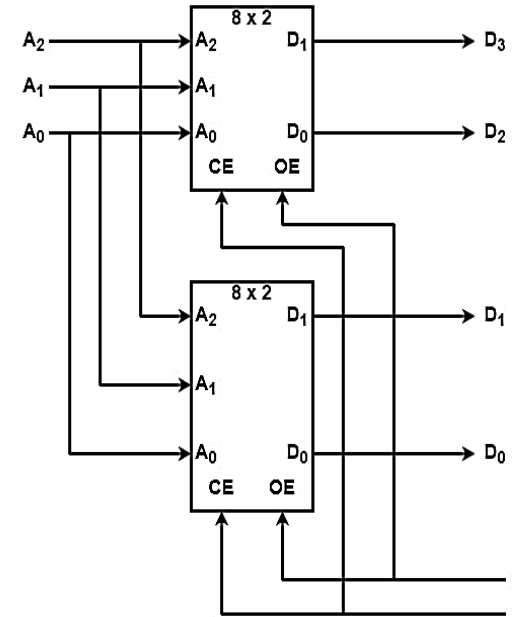


2.5 Memory Sub system Configuration

- There is technique for joining memory chips to form a memory subsystem.
- Two or more chips can be combined to generate a memory with more bits per location.
- This is done by linking the corresponding address and control signals of the chips and linking their data pins to various bits of the data bus.

Construction of 8 x 4 ROM from two 8 x 2 ROMs:

- **Two 8 x 2 chips** can be **combined** to generate an **8 x 4 memory** as displayed in the figure.
- **Both chips get the equal three address inputs** from the bus, and the same chip enable (CE) and output enable (OE) signals.
- The data chips of the first chips are linked to bits 3 and 2 of the data bus (i.e. D_3 and D_2)
- and those of the other chips are linked to bits 1 and 0 (i.e. D_1 and D_0)
- When the CPU reads data, it locates the address on the address bus.
- Both chips read in address bits A_2 , A_1 , and A_0 and implement their internal decoding..
- If the **CE** (Chip Enable) and **OE** (Output Enable) signals are **activated**, the **chips output their data onto the four bits** of the data bus.
- Because the address and enable signals are equal for both chips, **either both chips and neither chip is active** at any given time.
- The device never has only one of the two, active.
- For this reason, they facilitate as only a single 8 x 4 chip at least until the CPU is concerned.



Construction of 16 x 2 ROM from two 8 x 2 ROM chip:

- Instead of generating wider words, chips can be combined to make more words.
- The similar two 8 x 2 chips could alternatively be configured as a 16 x 2 memory subsystem.

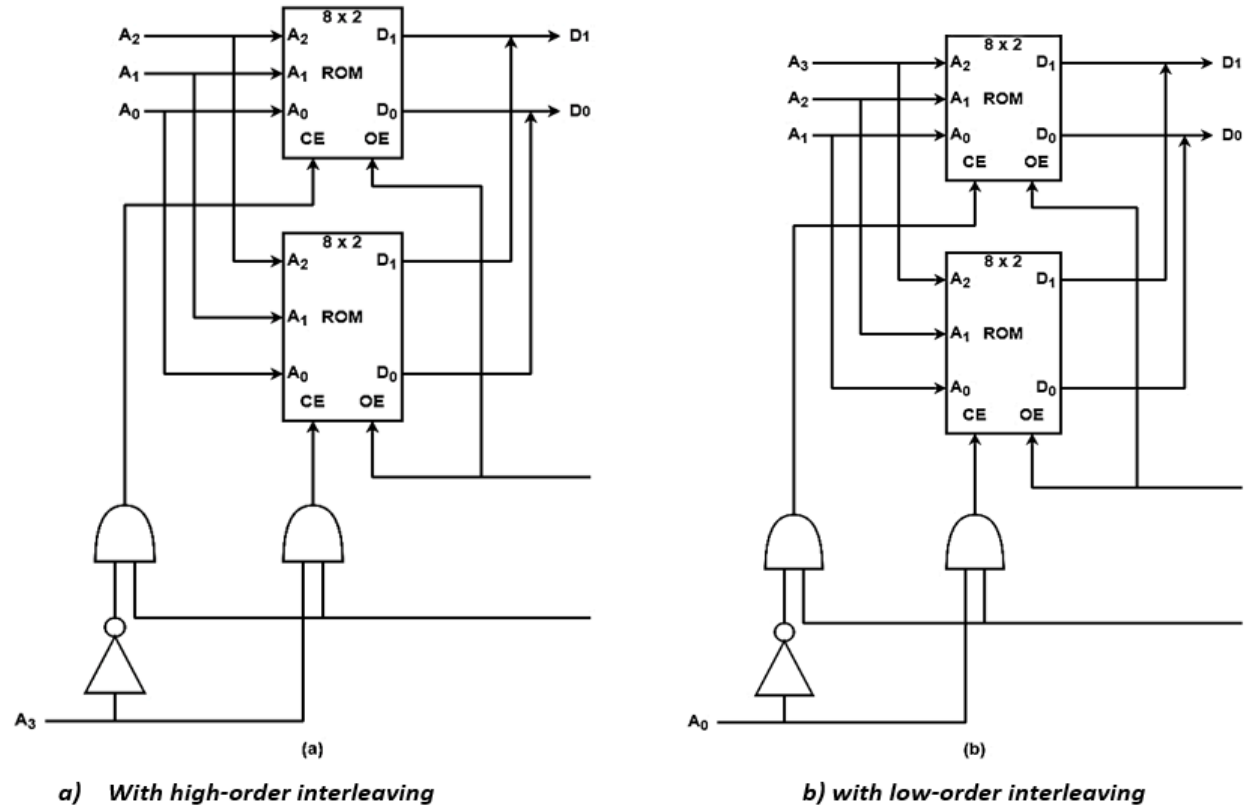
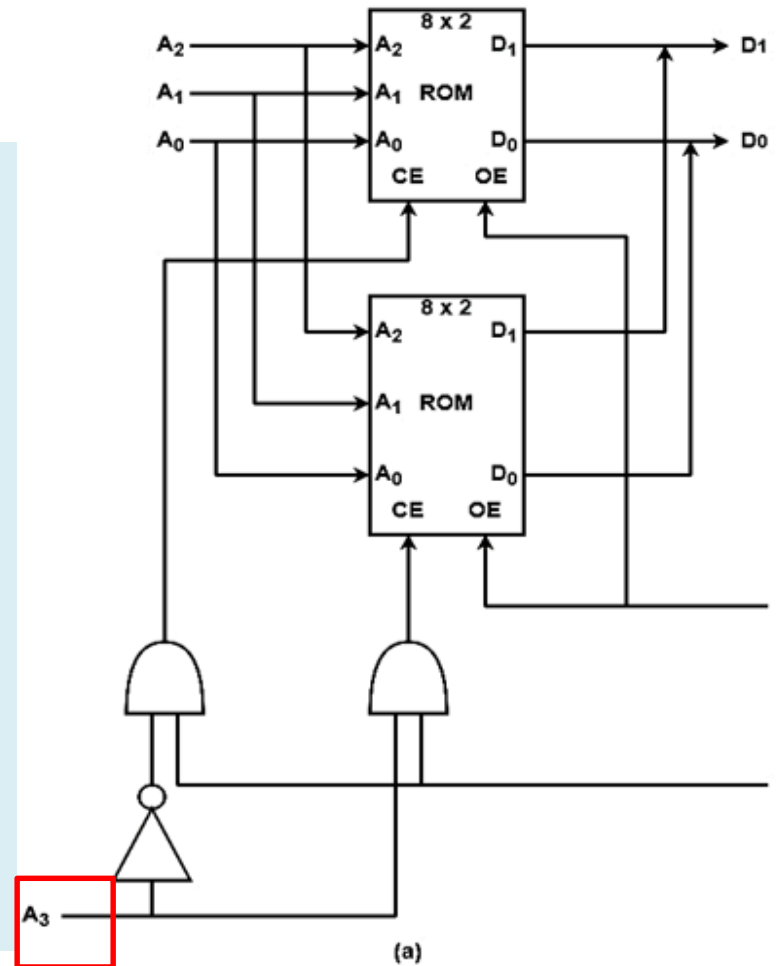


Figure: A 16 x 2 memory subsystem constructed from two 8 x 2 ROM chip
Microprocessor and Computer
Architecture, BSE-II

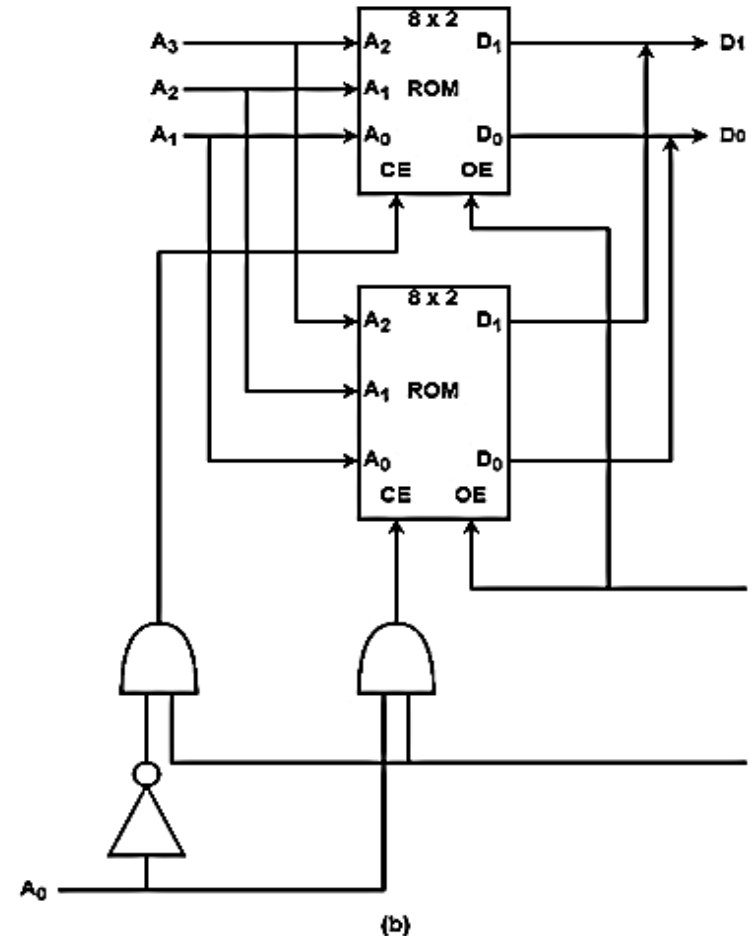
i) Using high-order interleaving

- The configuration in figure (a) uses **high-order interleaving**. All memory locations inside a chip are contiguous within system memory.
- The **upper chip** is configured as memory **locations 0 to 7** (0000 to 0111) and
- the **lower chip** as **locations 8 to 15** (1000 to 1111).
- The upper chip **always** has **$A_3=0$** and the lower chip has **$A_3=1$** .
- This difference can choose one of the two chips.
- When **$A_3=0$** , the **upper chip is enabled** and the **lower chip is disabled**.
- The output enables can be linked, because only the chip that is enabled with output data.
- Both chips correspond to similar data bits, both are linked to D_1 and D_0 of the data bus



ii) Using low-order interleaving

- Consider the configuration displayed in figure (b) which uses low-order interleaving.
- The **upper chip is enabled** when $A_0=0$, or by addresses XXX0, in this case, **0, 2, 4, 6, 8, 10, 12, and 14**.
- The **lower chip is enabled** when $A_0=1$, which is correct for addresses **1, 3, 5, 7, 9, 11, 13, and 15**.

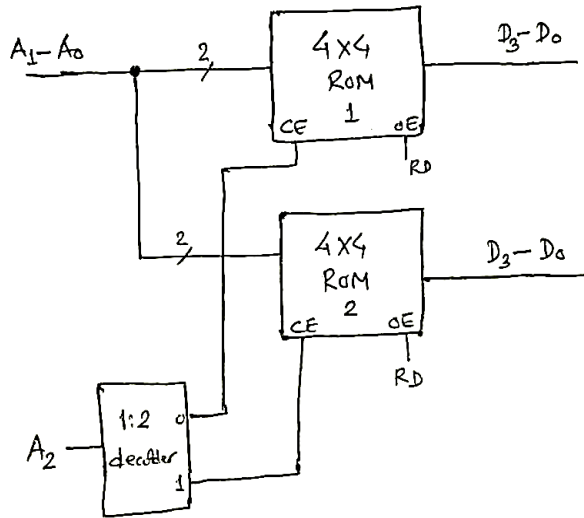


Classwork:

1. **Design an 8x4 ROM using 4x4 ROM chips using higher order interleaving. Also explain its operation.**
2. **Design a 16x4 memory subsystem with low order interleaving using 8x2 memory chips for a computer system with an 8-bit address bus.**
3. **Design a 32x 8 memory subsystem with high-order interleaving using 16x2 memory chips for a computer system with an 8-bit address bus.**

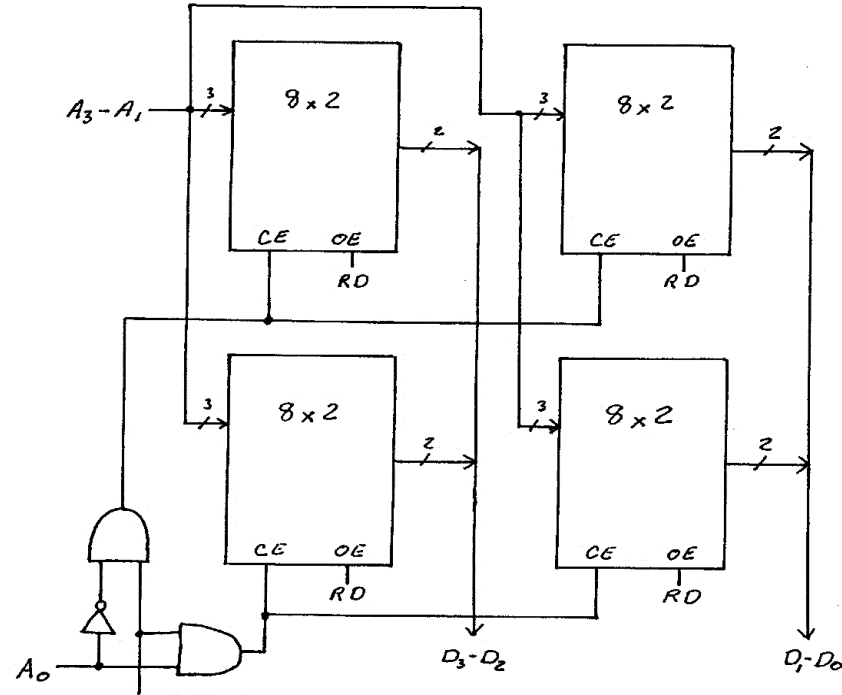
1. Design an 8x4 ROM using 4x4 ROM chips using higher order interleaving. Also explain its operation.

- Here:
- Available ROM size = 4x4
- Required ROM size = 8x4
- No. of required ROMs = $\frac{8 \times 4}{4 \times 4} = 2$
- Since, the multiplicand is different, we will have vertical arrangement of ROMs as below:



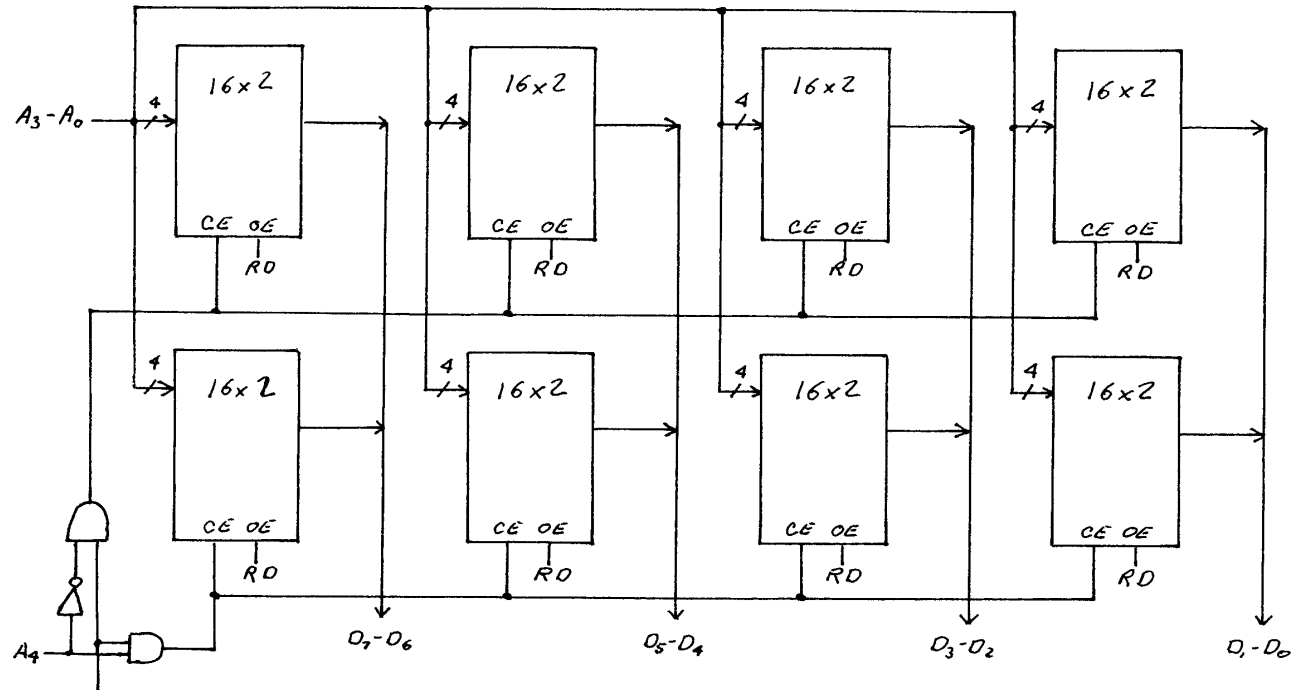
- When A₂= 0, the 1:2 decoder outputs form the first line (i.e 0) and thus enables ROM1
- When A₂= 1, the 1:2 decoder outputs form the second line (i.e 1) and thus enables ROM2.
- When address line (A₂-A₀) is from 000 to 011, the upper ROM1 chip is enabled and the corresponding data (D₃-D₀) will be out.
- When address line (A₂-A₀) is from 100 to 111, the upper ROM1 chip is enabled and the corresponding data (D₃-D₀) will be out.

2. Design a 16x4 memory subsystem with low order interleaving using 8x2 memory chips for a computer system with an 8-bit address bus.



Note: Decoder can be used for bit interleaving as in qstn 1

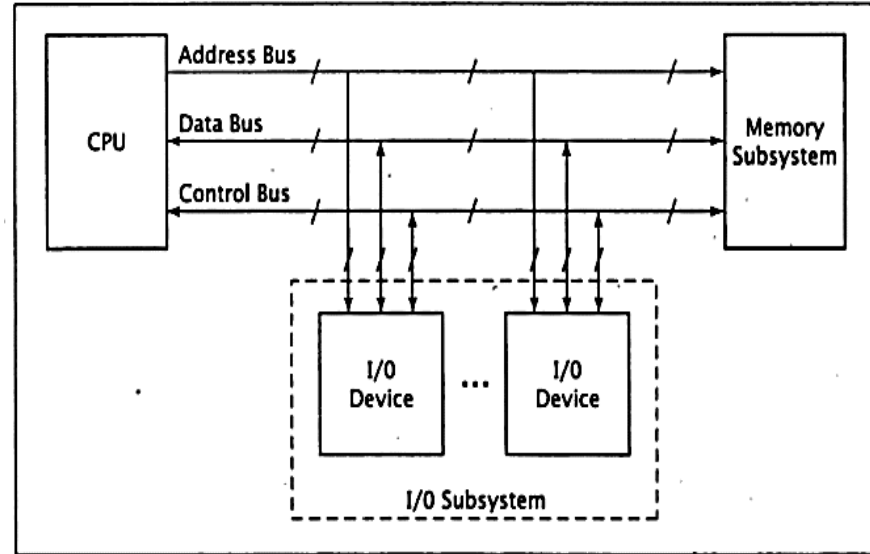
3. Design a 32x 8 memory subsystem with high-order interleaving using 16x2 memory chips for a computer system with an 8-bit address bus.



Note: Decoder can be used for bit interleaving as in qstn 1

6. I/O subsystem Organization and Interfacing

- The I/O subsystem is treated as an independent unit in the computer.
- The CPU initiates I/O command generally as Read, Write, Scan, etc.
- As shown in figure:
 - The **I/O device is connected to the computer system address, data and control buses.**
 - **Each I/O device includes I/O circuitry** that interacts with the buses.



i) Input device

- The data from the input device goes to the tri-state buffer.
- **When the values on the address bus and control bus are correct, the buffers are enabled and data passes onto the data bus.**
- The CPU, then can read these data.
- The key to this design is the enable logic.
- For the input device, the read signal (RD) should be asserted.
- Figure (b) shows the enable logic for an input device at address 11110000 with 8-bit address and control signals RD and $IO/\bar{M}$ (IO/Memory).

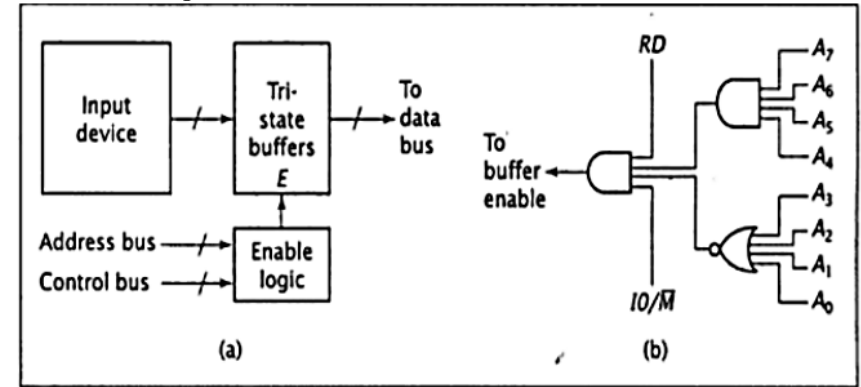


Figure: An Input device: a) with its interface, and b) the enable logic for the tri-state buffers

ii) Output Device:

- For the interface circuit for an output device, the tri-state buffer is replaced by a register.
- Only the device with the correct address will read it in.**
- The Load logic (LD) plays the role of the enable logic as in the input device.
- When this logic receives the correct address and control signals, it asserts the LD signal of the register, causing it to read data from the system's data bus.**
- The output device can then read the data from the register** at its leisure while the CPU performs other tasks.
- The logic to generate the load signal for an output device at address 1111 0000 is shown in figure (b).

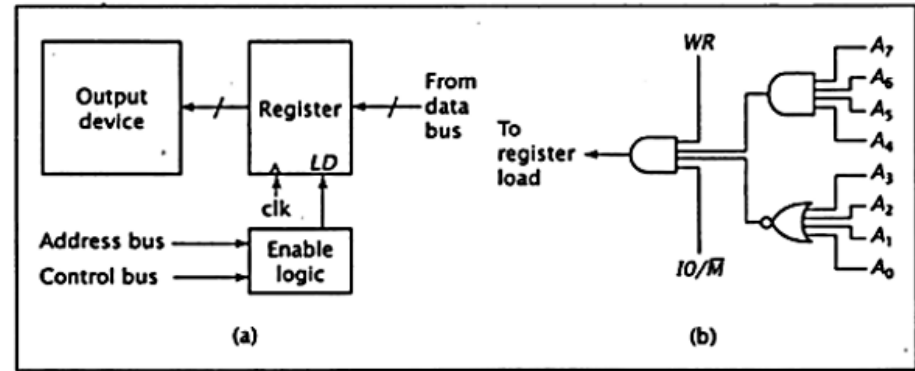


Figure: An Output device: a) with its interface and ,b) the load logic for the register

iii) Bidirectional IO device

- Some devices are used for both input and output. A personal computer's hard disk drive falls into this category.
- Such a device requires a combined interface that is essentially two interfaces, one for input and the other for output.
- Some logic elements, such as the gates that check the address and the address bus, can be used to generate both the buffer enable and register load signals.
- Figure below shows a combined I/O interface for address 1111 0000.

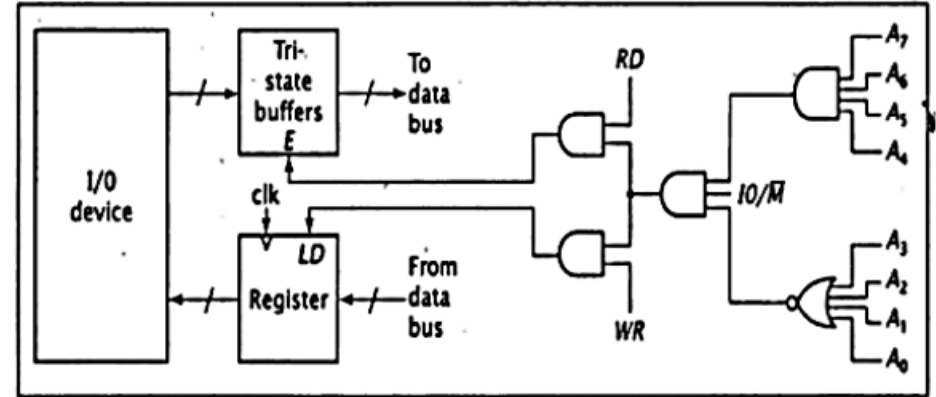
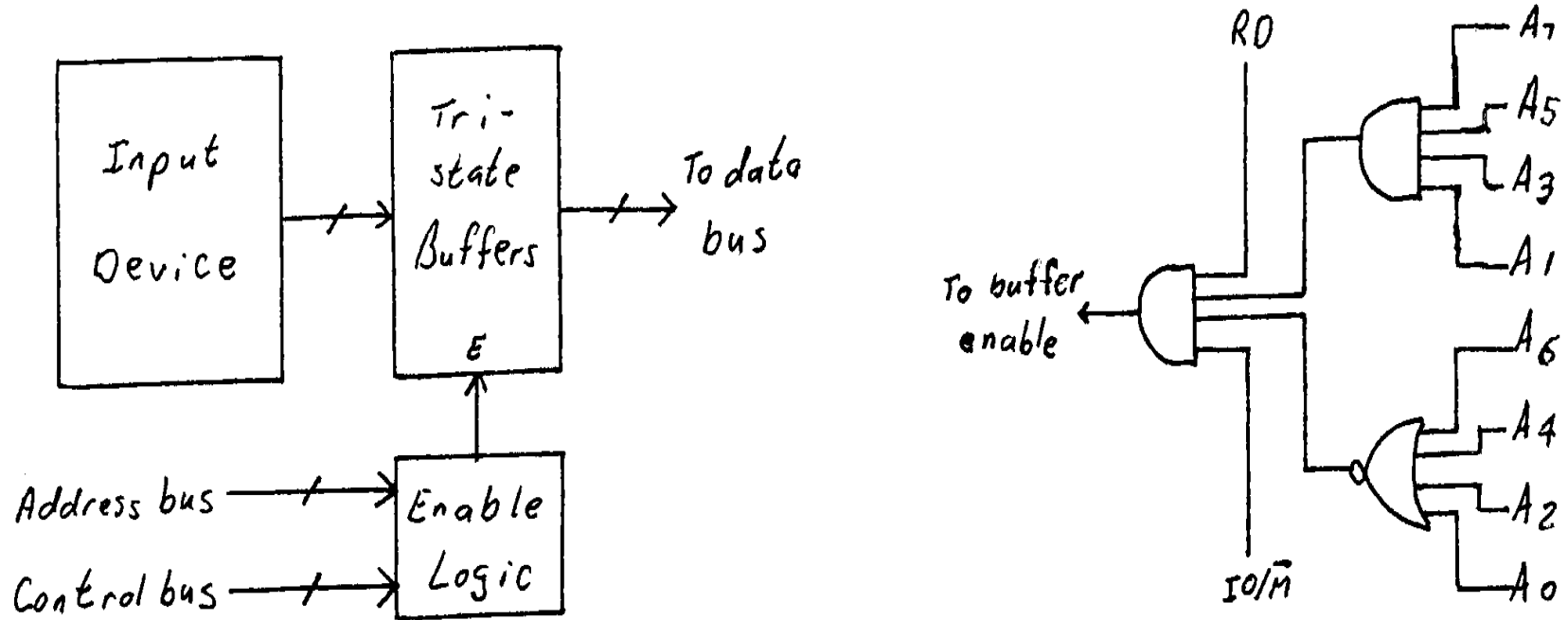
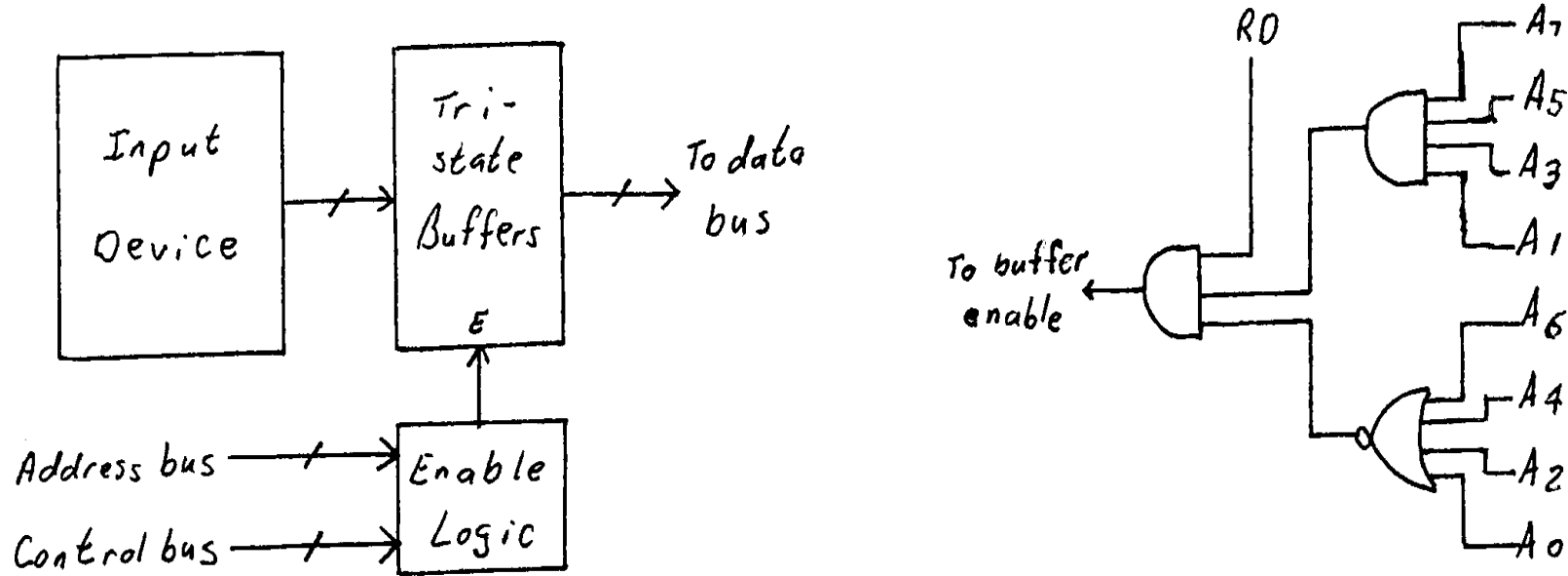


Figure: An bidirectional Input/Output device with its interface and enable/load logic

1. Design an interface for an input device which has binary address 1010 1010. Its computer system uses isolated I/O.



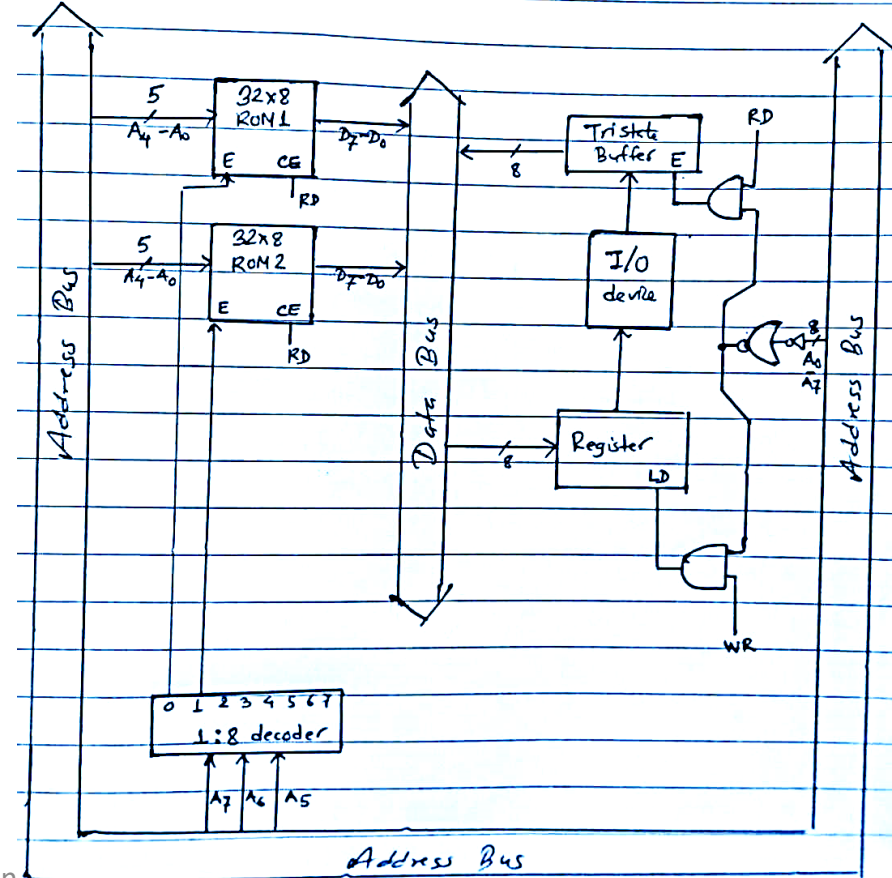
2. Design an interface for an input device which has binary address 1010 1010. Its computer system uses memory mapped I/O.



6. A CPU with 8-bit data bus and 8-bit address bus uses memory mapped I/O. It has 32 byte of ROM at 00H and 32 byte of ROM at 20H. It also has an I/O device at FFH. Show the load logic and enable logic.

- Here:
- Data bus width = 8 bits (D7-D0)
- Address bus width = 8 bits (A7-A0)
- ROM1 size= 32x8 starting at 00H (5 Address lines: A4-A0)
- ROM2 size= 32x8 starting at 20H (5 Address lines: A4-A0)
- I/O device starting at FFH

Memory	HexCode	A7	A6	A5	A4	A3	A2	A1	A0
ROM1	00H	0	0	0	0	0	0	0	0
	1FH	0	0	0	1	1	1	1	1
ROM2	20H	0	0	1	0	0	0	0	0
	3FH	0	0	1	1	1	1	1	1
IO	FFH	1	1	1	1	1	1	1	1

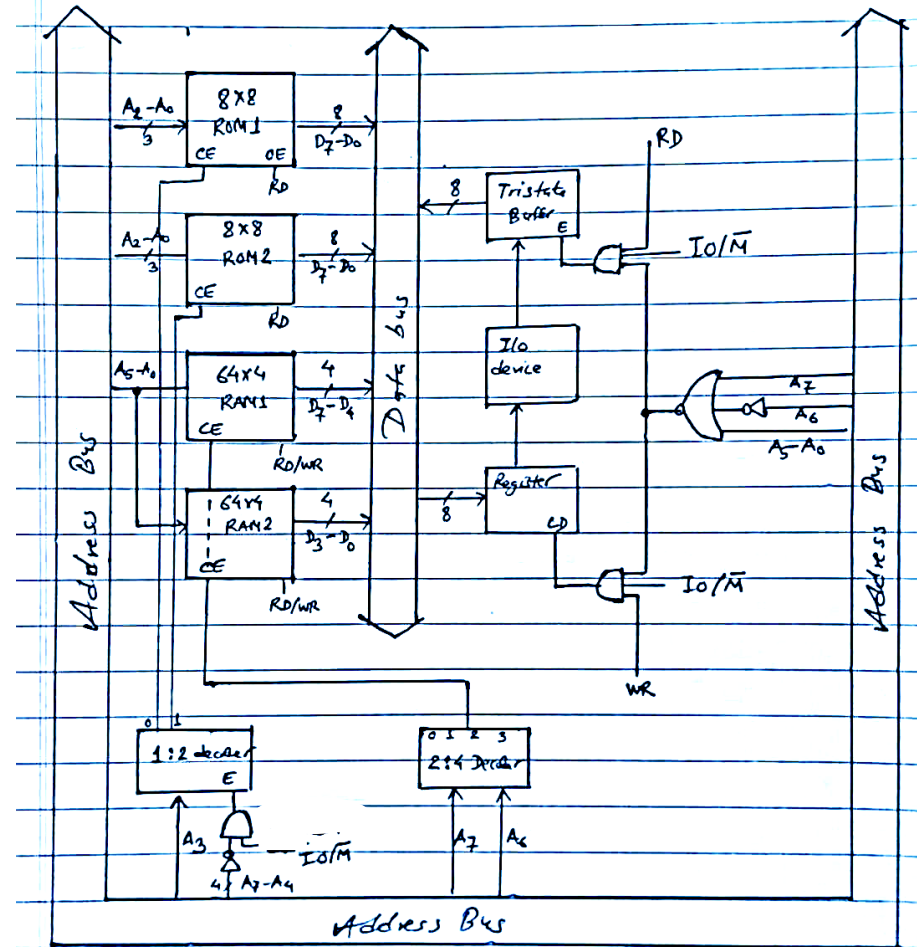


7. A computer system with an 8-bit address and an 8-bit data bus using isolated I/O. It has 16 x 8 ROM starting at address 00H constructed using 8 x 8 chips; 64 x 8 of RAM starting at address 80H constructed using 64 x 4 chips. There is an I/O device at 40H. Show the design for the system.

Here:

- Data bus width = 8 bits (D7-D0)
- Address bus width = 8 bits (A7-A0)
- ROM1 size= 8x8 starting at 00H (Address lines: A2-A0)
- ROM2 size= 8x8
- RAM1 size= 64x4 starting at 80H (Address lines: A4-A0)
- RAM2 size= 64x4
- I/O device starting at 40H

Memory	HexCode	A7	A6	A5	A4	A3	A2	A1	A0
ROM1	00H	0	0	0	0	0	0	0	0
	07H	0	0	0	0	0	1	1	1
ROM2	08H	0	0	0	0	1	0	0	0
	0FH	0	0	0	0	1	1	1	1
RAM1 and RAM2	80H	1	0	0	0	0	0	0	0
	BFH	1	0	1	1	1	1	1	1
IO	40H	0	1	0	0	0	0	0	0

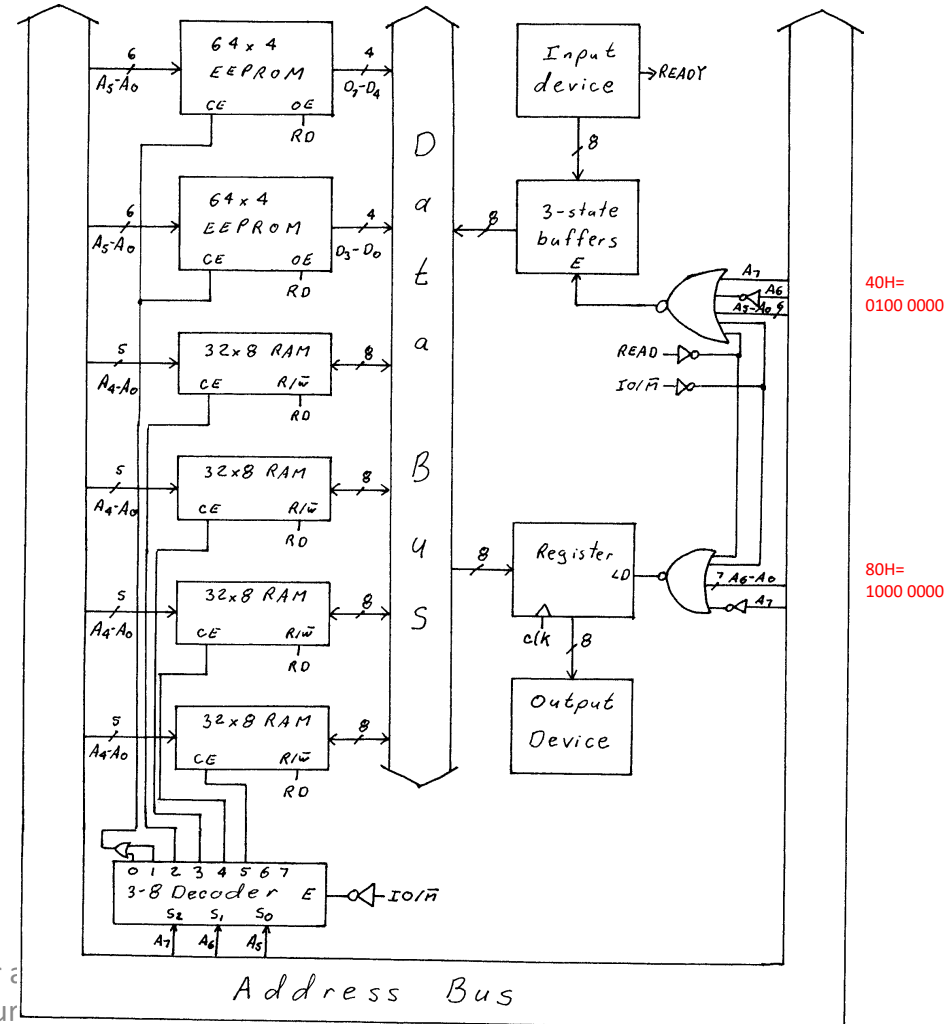


8. A computer system with 8-bit data bus and 8-bit control bus uses isolated I/O. It has 64 bytes of ROM starting at 00H constructed using 64 x 4 chips; 128 bytes of RAM, starting at address 40H constructed using 32 x 8 chips; an input device with READY signal at address 40H; and an output with no READY signal at address 80H. Show the design for this system. Include all enable and load logic.

Here:

- Data bus width = 8 bits (D7-D0)
- Address bus width = 8 bits (A7-A0)
- ROM size= 64x8 starting at 00H (5 Address lines: A4-A0)
 - We require two 64x4 ROMs for its construction
- RAM size= 128x8 starting at 40H (6 Address lines: A5-A0)
 - We require five 32x8 chips for its construction
- Input device with READY signal at 40H
- Output device with no READY signal at 80H

Memory	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	Address	Decimal Value
ROM	0	0	0	0	0	0	0	0	00H	0
1 and 2	0	0	1	1	1	1	1	1	3FH	63
RAM 1	0	1	0	0	0	0	0	0	40H	64
2	0	1	0	1	1	1	1	1	5FH	95
RAM 2	0	1	1	0	0	0	0	0	60H	96
3	0	1	1	1	1	1	1	1	7FH	127
RAM 3	1	0	0	0	0	0	0	0	80H	128
4	1	0	0	1	1	1	1	1	9FH	159
RAM 4	1	0	1	0	0	0	0	0	A0H	160
5	1	0	1	1	1	1	1	1	BFH	191



Exercise:

1. Differentiate between Microprocessor and Microcontroller.
2. Differentiate between Von-Neumann and Harvard Architecture with suitable block diagrams.
3. Explain the phases in instruction cycle.
4. What are the applications of Microprocessor? Briefly explain.
5. What are the applications of Microcontroller? Briefly explain.
6. Explain the basic computer organization.
7. Explain the linear organization of 8x2 ROM chip.
8. Explain the horizontal organization of 8x2 ROM chip.

9. Construct an 8x4 ROM from two 8x2 ROMs. Also explain it.
10. Construct an 8x4 ROM from two 8x2 ROMs using high-order interleaving. Also explain it.
11. Construct an 8x4 ROM from two 8x2 ROMs using low-order interleaving. Also explain it.
12. Design a 16x4 memory subsystem with high order interleaving using 8x2 memory chips for a computer system with an 8-bit address bus.
13. Design a 16x4 memory subsystem with low order interleaving using 8x2 memory chips for a computer system with an 8-bit address bus.
14. Design an interface for an input device which has binary address 1010 1010. Its computer system uses isolated I/O.
15. Design an interface for an input device which has binary address 1010 1010. Its computer system uses memory mapped I/O.
16. Design an interface for a bidirectional Input/output device that has binary address 1000 0001. Its computer system uses Isolated I/O.

17. A CPU with 8-bit data bus and 8-bit address bus uses memory mapped I/O. It has 32 byte of ROM at 00H and 32 byte of ROM at 20H. It also has an I/O device at FFH. Show the load logic and enable logic.
18. A computer system with an 8-bit address and an 8-bit data bus using isolated I/O. It has 16 x 8 ROM starting at address 00H constructed using 8 x8 chips; 64 x 8 of RAM starting at address 80H constructed using 64 x 4 chips. There is an I/O device at 40H. Show the design for the system.
19. 8. A computer system with 8-bit data bus and 8-bit control bus uses isolated I/O. It has 64 bytes of ROM starting at 00H constructed using 64 x 4 chips; 128 bytes of RAM, starting at address 40H constructed using 32 x 8 chips; an input device with READY signal at address 40H; and an output with no READY signal at address 80H. Show the design for this system. Include all enable and load logic.
20. A computer has a CPU with 8-bit address bus and 16-bit data bus. The computer uses Isolated I/O. It has 64 x 16 of ROM at 00 H; constructed using two 32 x 16 ROM chips. It also has 32 x16 of RAM at C0 H. The system has an input device at 17 H and an output device at B5 H. Show the design for the system including all the required logic. [PU 2018 Fall]
21. A computer system with an 8-bit address bus and an 8-bit data bus uses memory-mapped I/O. It has 32 bytes of ROM starting at address 00H constructed using a single 32x8 chip; 128 bytes of RAM starting at address 80H constructed using 64x4 chips; a bidirectional I/O device with no READY signal at address 40H; and an input device with no READY signal at address 60H. Show the design for this system. Include all enable and load logic.

End of Chapter